

Date: 2026-05-13
Request ID: 100048553

Hotplate		
Element(s)	Al, Co, Pt, Si, Ti	
SOP#		
Cocktail	None	
sample(s)	weight (g)	volume (mL)
200128003_1	40.1290	50
200128003_2	40.1538	50

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Al ICP analysis

sample	Dilution	conc. [mg/L]	Al_ppm	%Al
200128003_1	50/0.75	2.287	189.93	0.02
200128003_2	50/0.75	2.233	185.33	0.02

al result: 187.63 ppm

al result: 0.02 %

al oxide factor:

al oxide result: 0.00 %

al lot: *Manufacturer: Assurance, 27-171ALT exp: 2026-07-30*

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm	%Co
200128003_1	50/3	2.786	57.84	0.01
200128003_2	50/3	2.709	56.22	0.01

co result: 57.03 ppm

co result: 0.01 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm	%Pt
200128003_1	50/50	0.000	0.00	0.00
200128003_2	50/50	0.000	0.00	0.00

pt result: 0.00 ppm

pt result: 0.00 %

pt oxide factor:

pt oxide result: 0.00 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Si ICP analysis

sample	Dilution	conc. [mg/L]	Si_ppm	%Si
200128003_1	50/50	2.806	3.50	0.00
200128003_2	50/50	2.785	3.47	0.00

si result: 3.48 ppm

si result: 0.00 %

si oxide factor:

si oxide result: 0.00 %

si lot: *Manufacturer: Assurance, 28-87SIX exp: 2026-08-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm	%Ti
200128003_1	10/3	2.862	11.89	0.00
200128003_2	10/3	2.793	11.59	0.00

ti result: 11.74 ppm

ti result: 0.00 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s):

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$ppm\ M+ = [conc.][volume][dilution]/[weight]$$

$$\%M+ = [conc.][volume][dilution]/([weight]*10,000)$$

$$\%MO = [oxide\ factor]*\%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1-(\%LOI)/100])$$

$$\%M+ \ [dried] = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \ [dried] = \%MO / ([1-(\%LOI)/100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS])/([weight])$$

$$\%Cr_2O_3 = 1.462*[\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$